



Supporting Information for

The hidden order of Turing patterns in arid vegetation ecosystems

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Supporting Information Text

S1. Other mathematical models. In addition to the Rietkerk model Eq. (1) in the main text, we describe other reaction-diffusion models widely used in studying regular vegetation patterns of arid and semi-arid climates.

S1.1 The generalized Klausmeier model. Klausmeier (1999) proposed a two-equation model to describe the formation of vegetation stripes in sloping landscapes (1). Subsequently, numerous studies have generalized the Klausmeier model to flat surfaces (2–4). Here, we adopt the generalized Klausmeier model with a nondimensionalized form:

$$\begin{cases} \frac{\partial u}{\partial t} = wu^2 - Bu + \Delta u \\ \frac{\partial w}{\partial t} = A - w - wu^2 + D_w \Delta w \end{cases}, \quad [S1]$$

where, u is plant biomass, w is water, A is rainfall, B is plant mortality, D is the diffusion coefficient of water. wu^2 indicates that plants can enhance water infiltration.

S1.2 The Hardenberg model. Hardenberg et al. (2001) proposed a two-equation model to study desertification (5), and we adopt the nondimensionalized form of it:

$$\begin{cases} \frac{\partial n}{\partial t} = \frac{\gamma w}{1+\sigma w} n - n^2 - \mu n + \Delta n \\ \frac{\partial w}{\partial t} = p - (1 - \rho n)w - w^2 n + \delta \Delta(w - \beta n) \end{cases}, \quad [S2]$$

where, n is plant biomass, w is groundwater, p is rainfall, $(1 - \rho n)$ means plants can reduce evaporation by shading, $w^2 n$ represents the local uptake of water by plants, $\Delta(w - \beta n)$ is derived from Darcy's law, $\frac{\gamma w}{1+\sigma w} n$ means the growth rate of the plant increases linearly with w for dry soil, n^2 represents local intraspecific competition, δ is the diffusion coefficient.

S1.3 A simplified model based on the Gilad model. Getzin et al. (2016) proposed a three-equation model to explain the fairy circle discovered in Australia (6), and this model is a simplified version modified from Gilad et al. (2007) (7). We adopt the nondimensionalized form of it:

$$\begin{cases} \frac{\partial v}{\partial t} = s(1 + \eta v)^2 v(1 - v) - v + \Delta b \\ \frac{\partial s}{\partial t} = \chi \frac{v+qf}{v+q} h - \frac{v_s}{1+R_s v} s - \gamma v(1 + \eta v)^2 s + \delta_s \Delta s, \\ \frac{\partial h}{\partial t} = r - \chi \frac{v+qf}{v+q} h - \frac{v_h}{1+R_h v} h + \delta_h \Delta(h^2) \end{cases}, \quad [S3]$$

where, v is plant biomass, s is soil water, h is surface water level. r is rainfall, η is root's augmentation, χ is infiltration rate in fully vegetated soil, q is biomass reference value beyond which infiltration rate under a patch approaches its maximum, γ is soil water consumption rate, v_s is soil water evaporation rate, v_h is surface water evaporation, R_s is soil water evaporation reduction due to shading, R_h is surface water evaporation reduction due to shading, f is infiltration contrast between bare soil and vegetated soil, δ_s is the transport coefficient for soil water, δ_h is the bottom friction coefficient between surface water and ground surface.

S1.4 The stochastic Rietkerk model. Dakos et al. (2011) used the stochastic Rietkerk model (Eq. (1) in the main text) to study the early warning signals of spatially patterned ecosystems (8). We adopt the model with Ito calculus:

$$\begin{cases} dO = \left(R - \alpha \frac{P+k_2 W_0}{P+k_2} O + D_O \Delta O \right) dt + \sigma_O d\eta(t) \\ \frac{\partial W}{\partial t} = \left(\alpha \frac{P+k_2 W_0}{P+k_2} O - \frac{g_{max} W}{W+k_1} P - r_w W + D_W \Delta W \right) dt + \sigma_W d\eta(t), \\ \frac{\partial P}{\partial t} = \left(c \frac{g_{max} W}{W+k_1} P - dP + D_P \Delta P \right) dt + \sigma_P d\eta(t) \end{cases}, \quad [S4]$$

where, $d\eta(t)$ is the Gaussian white noise with a standard deviation of σ_0 , the other parameters are the same as those in Eq. (1) in the main text.

S2. Numerical simulations. We numerically simulate Eqs. (S1)-(S3) and Eq. (S4) in Python 3.8 (pyopencl) and MATLAB 2022a. All simulations take a square domain L^2 with $L = 4096$. For Eqs.

(S1) – (S3), we use forward difference and two-order center schemes for the time derivative and Laplace operators, respectively. For Eq. (S4), we use the semi-implicit Euler–Maruyama and spectral Galerkin methods for the time derivative and Laplace operators, respectively (9). We use periodic boundaries to minimize boundary effects. Simulations are accelerated by the graphics processing units (NVIDIA Quadro K6000, Quadro P4000, Quadro P5000, Quadro P6000, Tesla K40c, GeForce RTX 3060, GeForce RTX 3090, and A100). Similar to the Rietkerk model, Eq. (1) in the main text, we apply two initial conditions:

[1] we randomly select 10% grid points to set fixed plant biomass and set plant biomass of other grid points to 0; we set all grid points to a fixed water content.

[2] in numerical disturbance experiments, we apply disturbances to the patterned system states obtained from the previous numerical simulations, and then we use the disturbed system states as the initial values.

We use almost the same parameter settings as the previous literature (4–6), and the details of parameter settings are shown below:

S2.1 The generalized Klausmeier model, Eq. (S1). The spatial resolution is 0.5, and the temporal resolution is 0.0001, $B = 0.45$, $D_w = 500$.

S2.2 The Hardenberg model, Eq. (S2). The spatial resolution is 0.5, and the temporal resolution is 0.0005, $\gamma = 1.6$, $\sigma = 1.6$, $\mu = 0.2$, $\rho = 1.5$, $\delta = 100$, $\beta = 3$.

S2.3 A simplified model based on the Gilad model, Eq. (S3). The spatial resolution is 1, and the temporal resolution is 0.00004, $q = 1.80$, $v_h = 2.25$, $v_s = 0.75$, $\chi = 60$, $\eta = 1.0$, $\gamma = 4.66$, $\eta = 1$, $\delta_s = 25$, $\delta_h = 2667$, $R_s = 0.3$, $R_h = 0.8$.

S2.4 The stochastic Rietkerk model, Eq. (S4). $\sigma_0 \in \{0, 0.1, 1\}$, $R = 0.7$, $t = 10,000$, the other parameters and initial condition are the same as the numerical simulation of Eq. (1) in the main text.

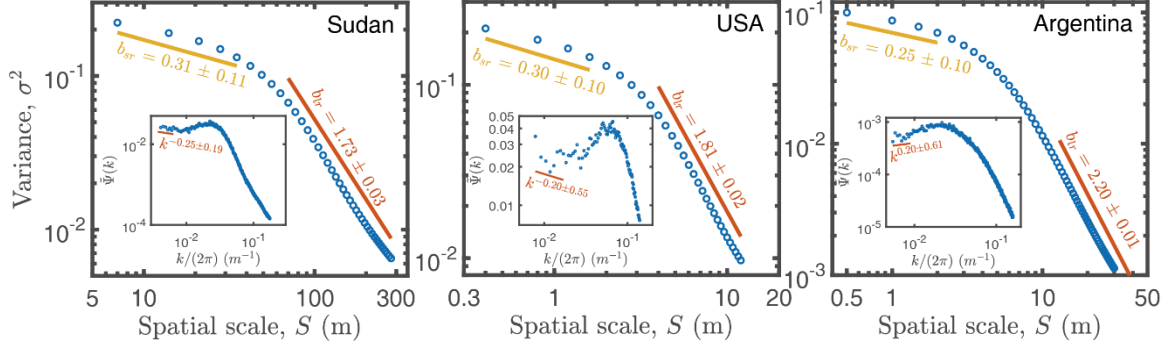


Fig. S1. The density fluctuation and circularly averaged spectral density (mean r-spectrum) of the vegetation patterns in Fig. 1A1-A3 of the main text. The vegetation pattern of Fig. 1A3 is in a hyperuniform state. The vegetation patterns of Fig. 1A1-A2 in the main text have spatial wavelengths (i.e., the presence of peaks in the circularly averaged spectral density) and spot vegetation patches, but are not in hyperuniform states. This spatial feature implies that the vegetation ecosystems in Fig. 1A1-A2 are in a low-resilience transient regime due to heterogeneous disturbances.

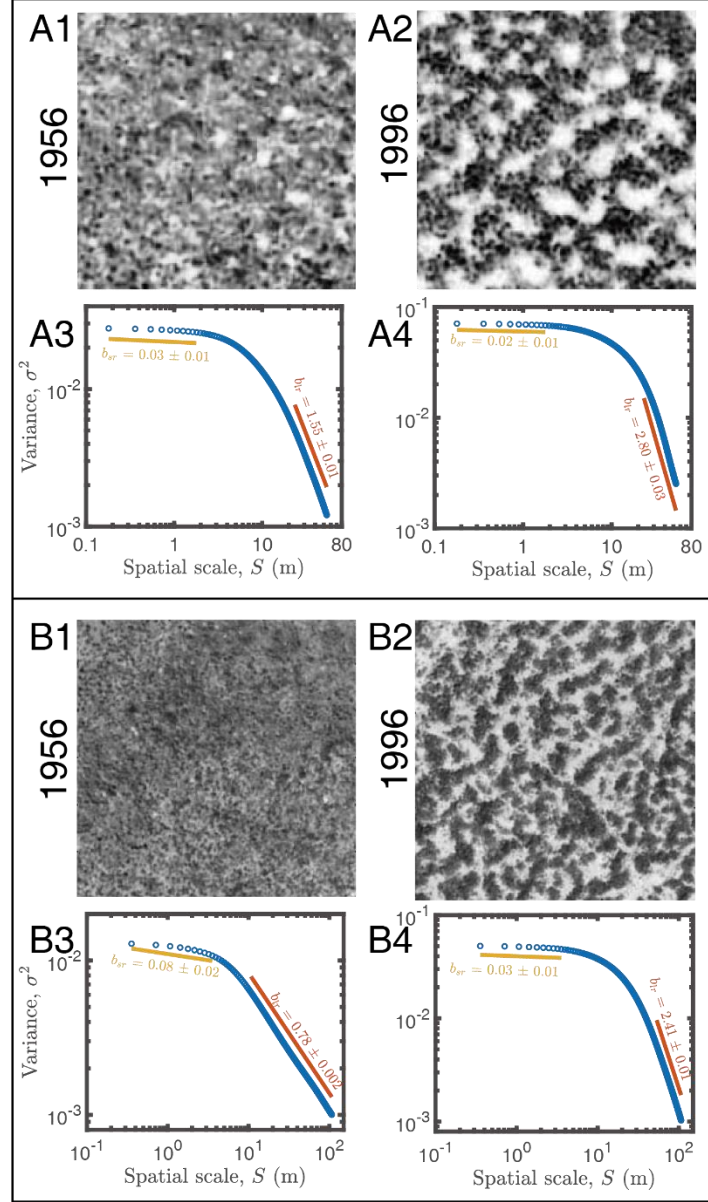


Fig. S2. The changes of either end of a 40-year interval in southern Niger. (A1) and (A2) are the 300 m by 300 m aerial photos of one area. Bright pixels correspond to bare soil, dark pixels correspond to woody vegetation, and intermediate greyscale pixels correspond to herbaceous vegetation. (A3) and (A4) are the density fluctuations of (A1) and (A2), respectively. (B1) and (B2) are the 600 m by 600 m aerial photos of another area. (B3) and (B4) are the density fluctuations of (B1) and (B2), respectively. With the formation of vegetation Turing patterns, the hyperuniform states (1996) developed from the spatial configuration of long-range aggregation (1956). These aerial photos cite from Barbier et al. (2006) (10).

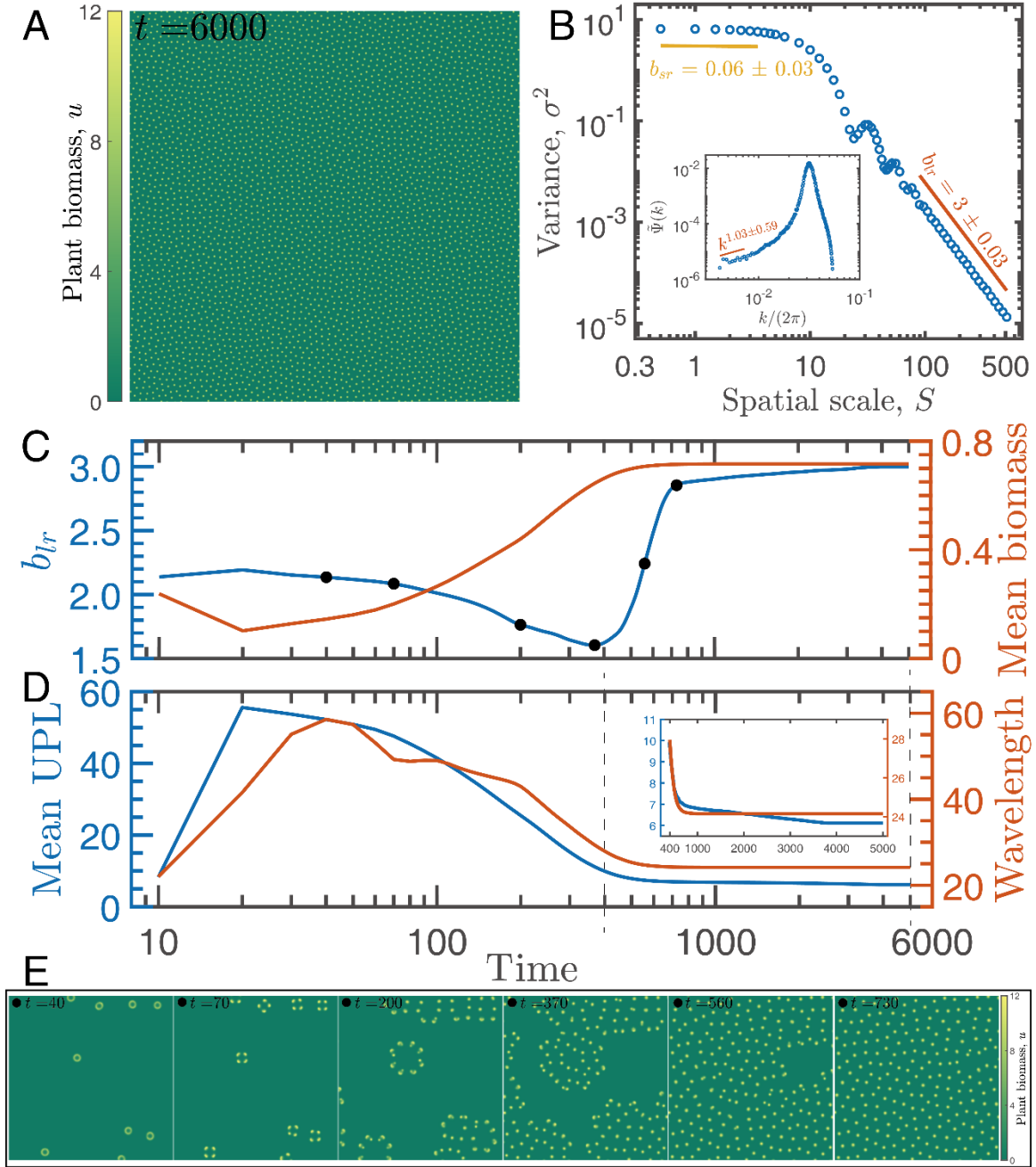


Fig. S3. The results of the generalized Klausmeier model, Eq. (S1). (A) The spatial pattern in the steady state. (B) The density fluctuation and circularly averaged spectral density (inset) of the spatial pattern (A). (C) The temporal changes of hyperuniform index and mean biomass. (D) The temporal changes of mean unchanneled path length (UPL) and spatial wavelength. Inset: the changes after $t = 400$. (E) The 1000x1000 slices from spatial patterns correspond to the black dots in (C). The typical transient behaviors are vegetation rings and self-replication, and the vegetation rings form in the early stage of evolution. The rainfall $A = 0.5$, $t = 5000$.

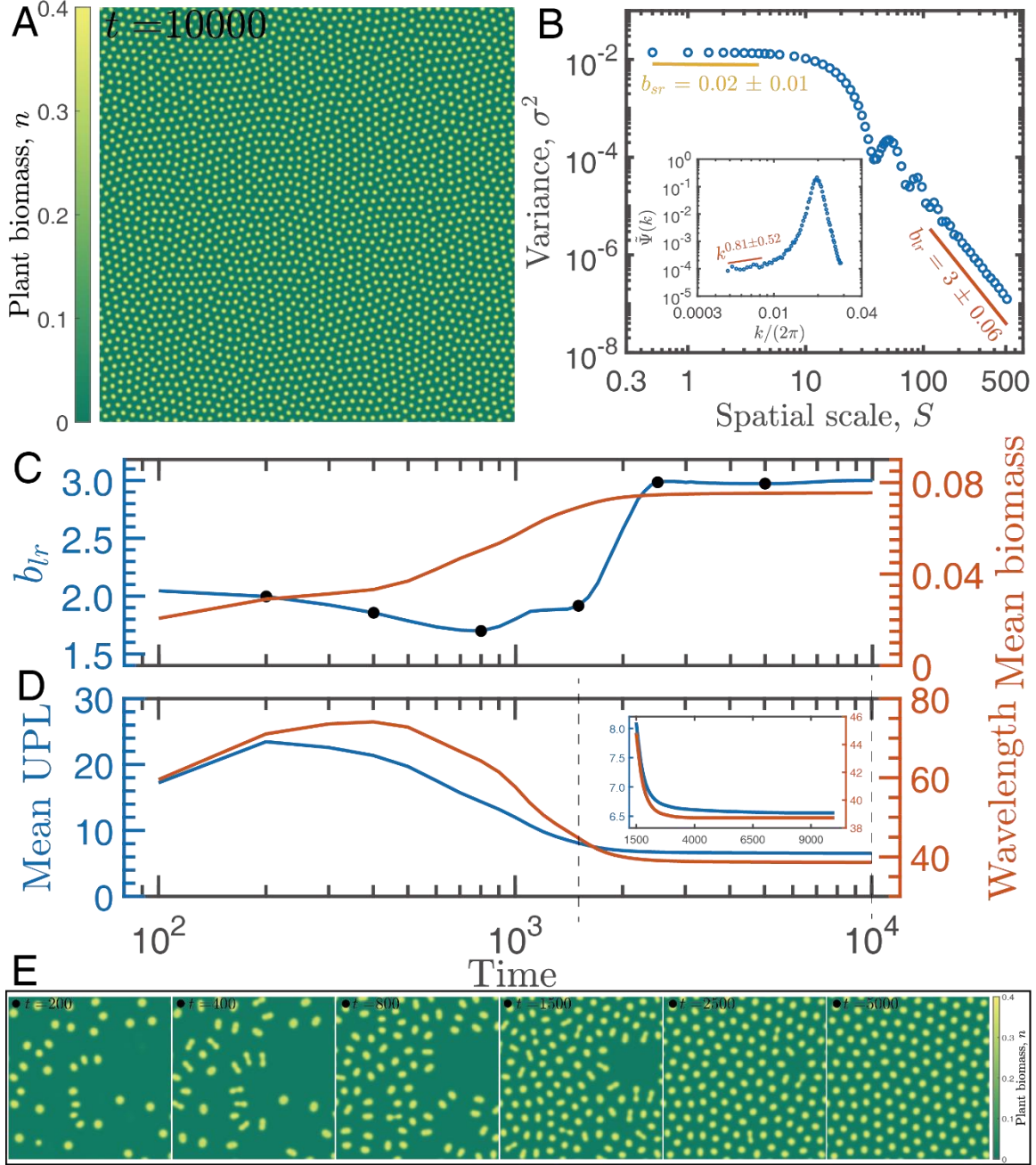


Fig. S4. The results of the Hardenberg model, Eq. (S2). (A) The spatial pattern in the steady state. (B) The density fluctuation and circularly averaged spectral density (inset) of the spatial pattern (A). (C) The temporal changes of hyperuniform index and mean biomass. (D) The temporal changes of mean unchanneled path length (UPL) and spatial wavelength. Inset: the changes after $t = 1500$. (E) The 1000x1000 slices from spatial patterns correspond to the black dots in (C). The typical transient behavior is vegetation self-replication. The rainfall $p = 0.15$, $t = 10000$.

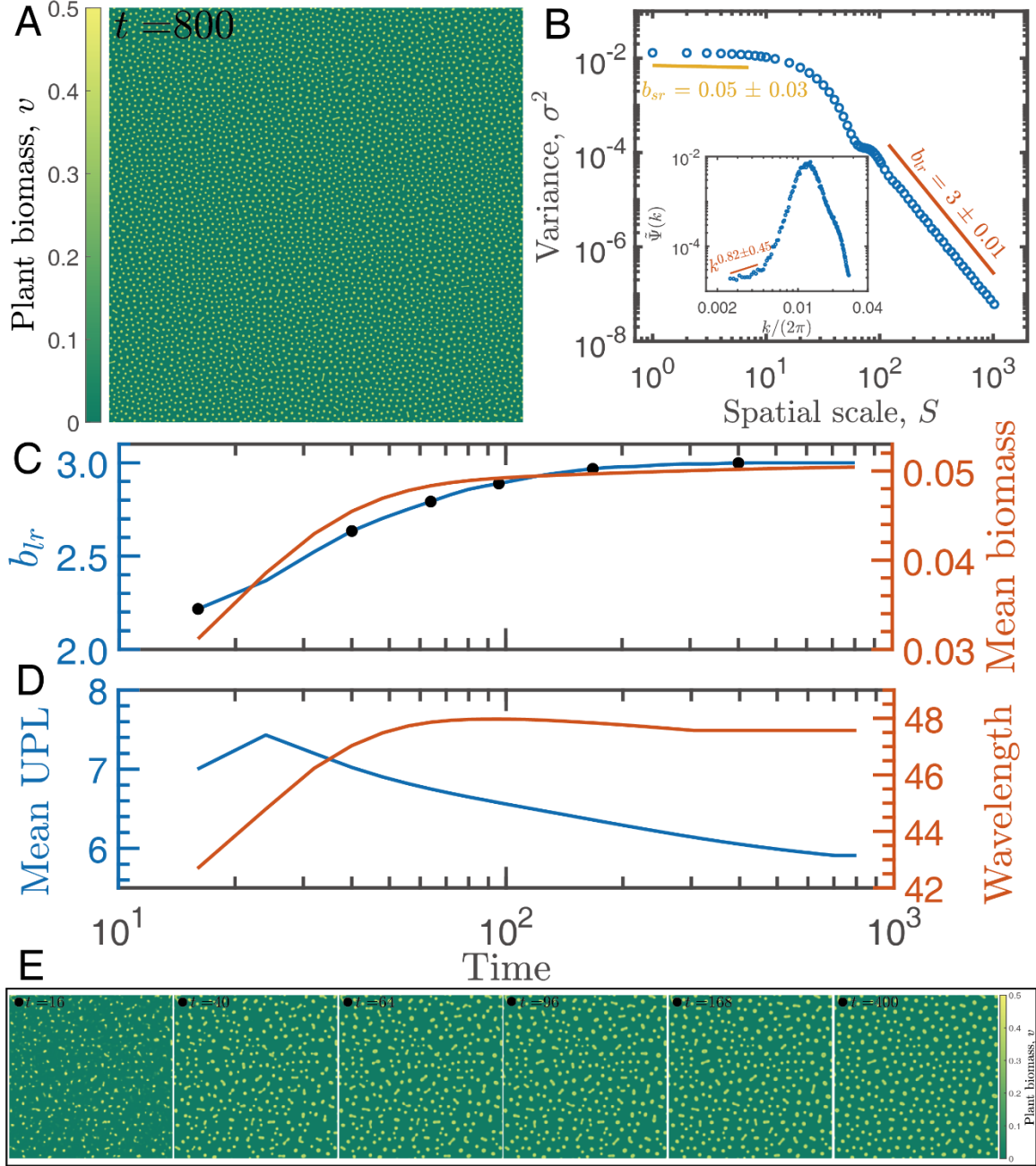


Fig. S5. The results of the Getzin model, Eq. (S3). (A) The spatial pattern in the steady state. (B) The density fluctuation and circularly averaged spectral density (inset) of the spatial pattern (A). (C) The temporal changes of hyperuniform index and mean biomass. (D) The temporal changes of mean unchanneled path length (UPL) and spatial wavelength. (E) The 1000x1000 slices from spatial patterns correspond to the black dots in (C). The typical transient behavior is vegetation ring. The rainfall $r = 1.6$, $t = 800$.

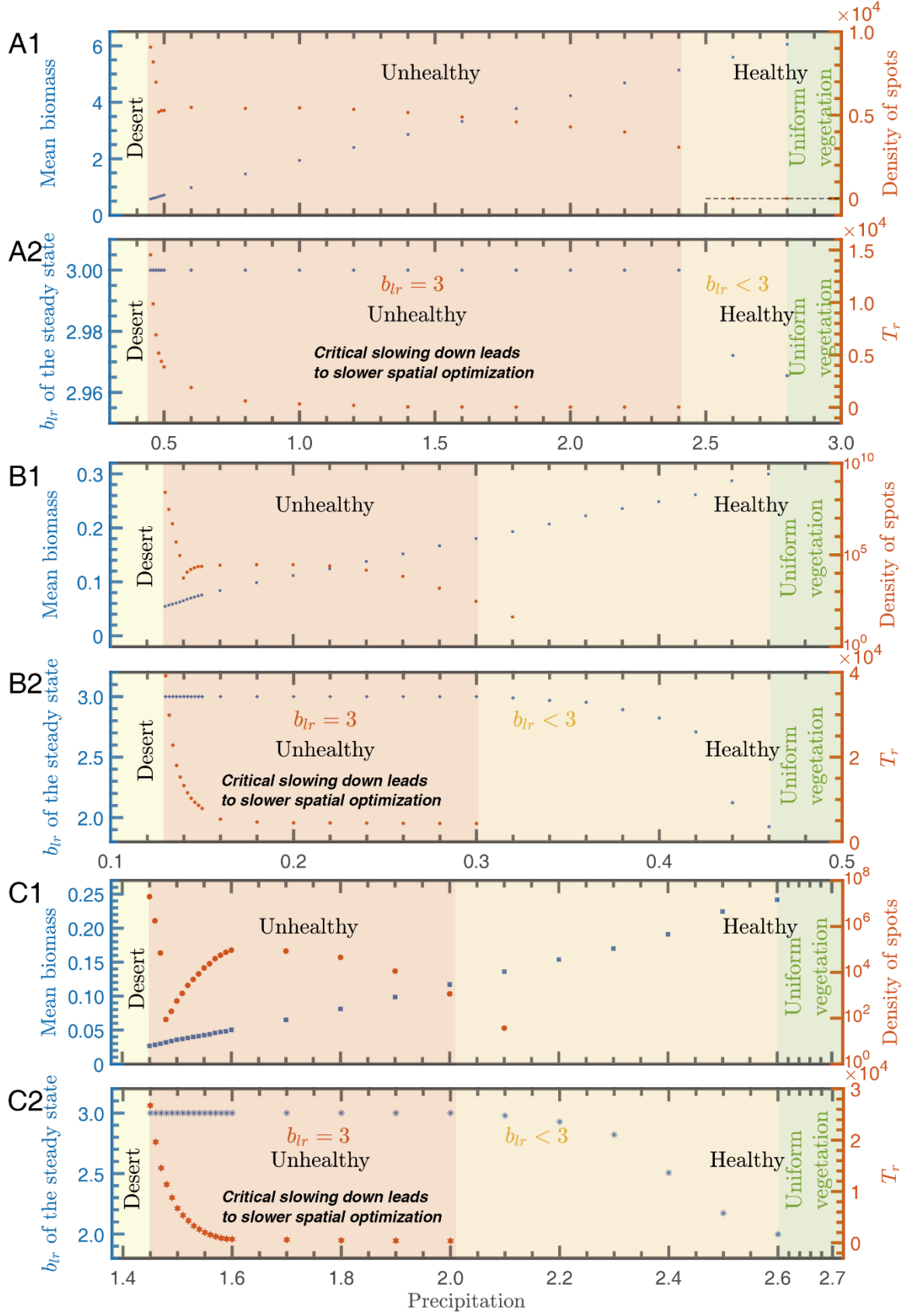


Fig. S6. The mean biomass, hyperuniform index, and the number of spots in the rainfall parameter space of the three models (Eq. (S1)-(S3)) in the steady states. A spot vegetation patch is defined as one with a circularity value exceeding 0.9. The density of spots is defined by the number of spot vegetation patches divided by the mean biomass. T_r represents the time required for vegetation to evolve to a spatial pattern with the hyperuniform index reaching 3. (A1)-(A2) The generalized Klausmeier model, Eq. (S1). We scan 13 evenly spaced values of A on the interval $[0.6, 3.0]$, and scan 7 evenly spaced values of A on the interval $[0.44, 0.5]$. The vegetation Turing patterns form between $A \in [0.45, 2.8]$, and the vegetation collapse at $A = 0.44$. (B1)-(B2) The Hardenberg model, Eq. (S2). We scan 17 evenly spaced values of p on the interval $[0.16, 0.48]$, and scan 12 evenly spaced values of p on the interval $[0.128, 0.15]$. The vegetation Turing patterns form between $p \in [0.13, 0.46]$, and the vegetation collapse at $p = 0.128$. (C1)-(C2) The Getzin model, Eq. (S3). We scan 12 evenly spaced values of r on the interval $[1.6, 2.8]$, and scan 16 evenly spaced values of r on the interval $[1.44, 1.59]$. The vegetation Turing patterns form between $r \in [1.45, 2.6]$, and the vegetation collapse at $r = 1.44$. These results are similar to Fig. 3 in the main text: the mean biomass decreases linearly with rainfall when Turing patterns form; the hyperuniform index reaches 3 after the spot vegetation patch appears; the density of spots increases rapidly with decreasing rainfall; T_r rises exponentially when the rainfall approaches the tipping point.

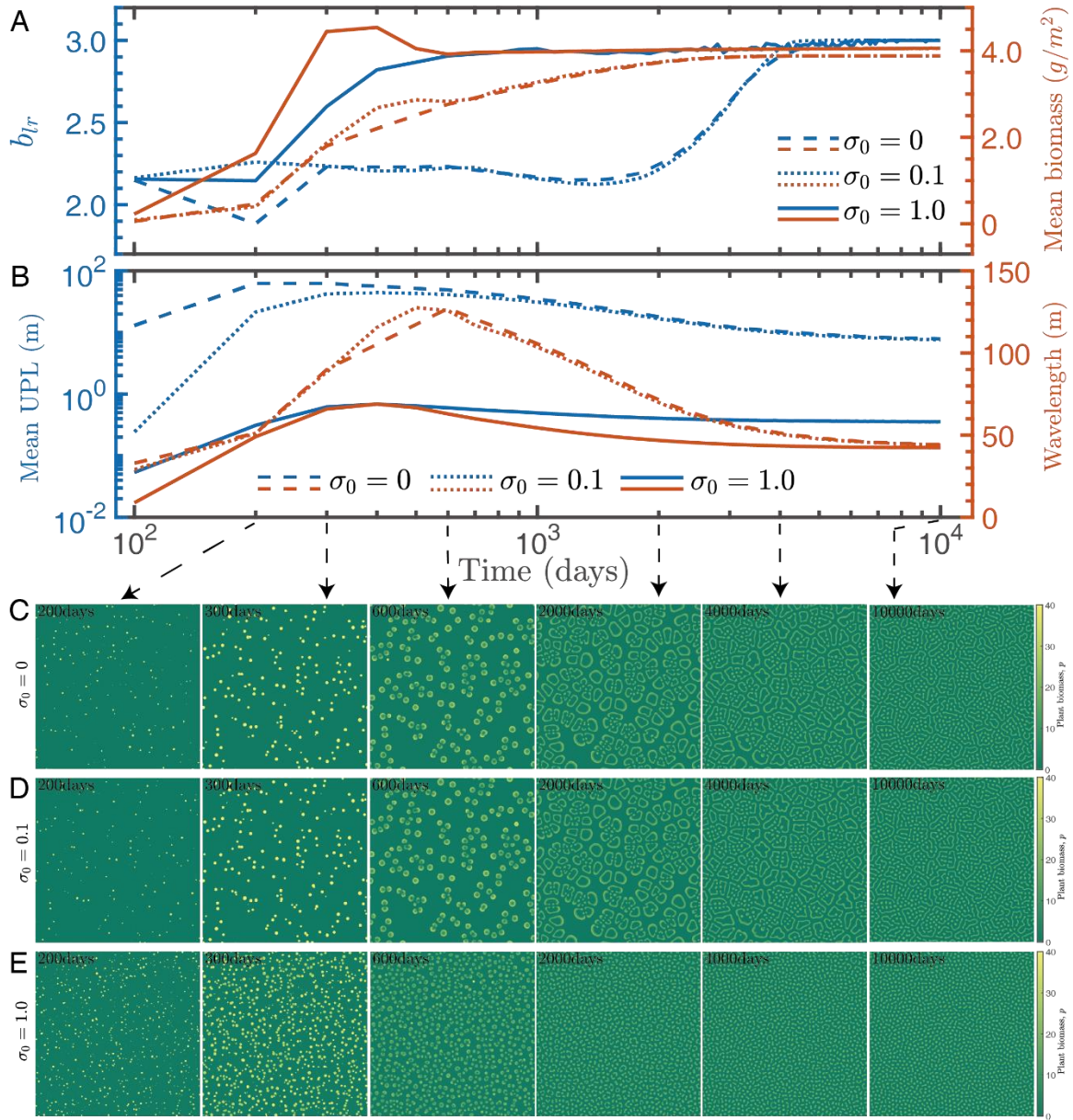


Fig. S7. The evolution of Eq. (S4) under different noise levels σ_0 . (A) The temporal changes of hyperuniform index and mean biomass. (B) The temporal changes of mean unchanneled path length (UPL) and spatial wavelength. Compared to the low noise level ($\sigma_0 = 0.1$) scenario, the mean UPL is remarkably short in the high noise level ($\sigma_0 = 1.0$) scenario. (C)-(E) The spatial patterns during evolution. The hyperuniform index reaches 3 at all noise levels in the steady states. While the low noise level ($\sigma_0 = 0.1$) slightly changes the transient behaviors of the Turing patterns, the high noise level ($\sigma_0 = 1.0$) significantly changes the transient behaviors of the Turing patterns (A). The high noise level causes the hyperuniform index and the mean biomass to reach high values quickly due to the better spatial optimization of water-use efficiency (B), which implies that stronger fluctuations favor rapid system evolution to high-resilience states. The rainfall $R = 0.7$, and the initial conditions of the simulations under different noise levels are identical to that of Fig. 4A in the main text.

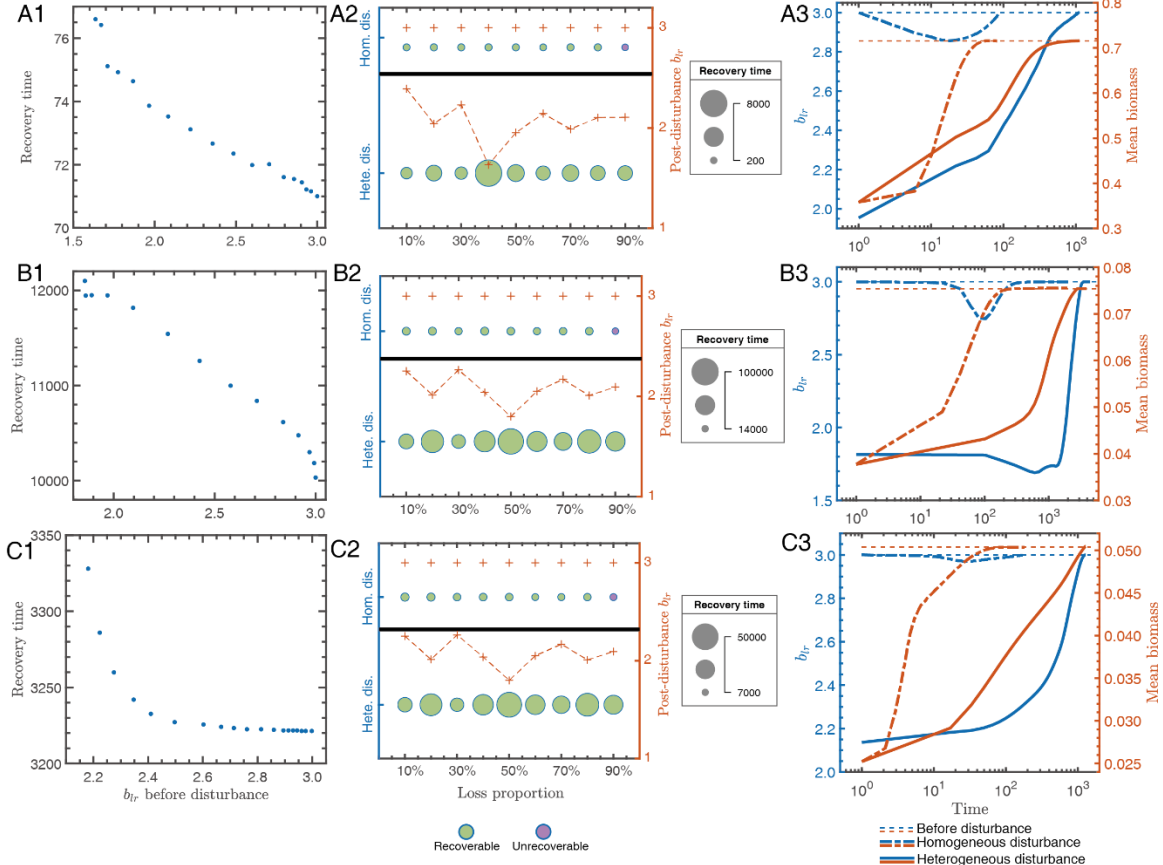


Fig. S8. The numerical disturbance experiments of the three models (Eq. (S1)-(S3)) under fixed rainfall. Similar to Fig.5 in the main text, we set two kinds of disturbance: homogeneous and heterogeneous (**Methods**). "Hom. dis." represents homogeneous disturbance, and "Hete. dis." represents heterogeneous disturbance. "Recovery time" is the days to recover a unit of mean biomass. (A1)-(A3) The generalized Klausmeier model, Eq. (S1), and $A = 0.5$. For (A2) and (A3), the disturbed spatial pattern is Fig. S4A. (B1)-(B3) The Hardenberg model, Eq. (S2) and $p = 0.15$. For (B2) and (B3), the disturbed spatial pattern is Fig. S5A. (C1)-(C3) The Getzin model, Eq. (S3), and $r = 1.6$. For (C2) and (C3), the disturbed spatial pattern is Fig. S6A. (A1), (B1), and (C1) Results for transient vegetation patterns under the homogeneous disturbance of 50% loss to the system state. (A2), (B2), and (C2) The recoverability and recovery time of vegetation under different types and intensities of disturbances. "Recoverable" means the mean biomass can recover to the pre-disturbance mean biomass. "Unrecoverable" means that vegetation collapse. (A3), (B3), and (C3) The temporal changes of hyperuniform index and mean biomass under the disturbance of 50% loss to the vegetation state. The recovery in (A2), (B2), and (C2) is the return of both mean biomass and hyperuniform index to the steady state, while the recovery in (A1), (B1), and (C1) is only the return of the mean biomass to the pre-disturbance state. These results are similar to Fig. 5 in the main text: a heterogeneous disturbance that destroys the hyperuniform structure increases the recovery time of the ecosystems compared to a homogeneous disturbance that maintains the hyperuniform structure.

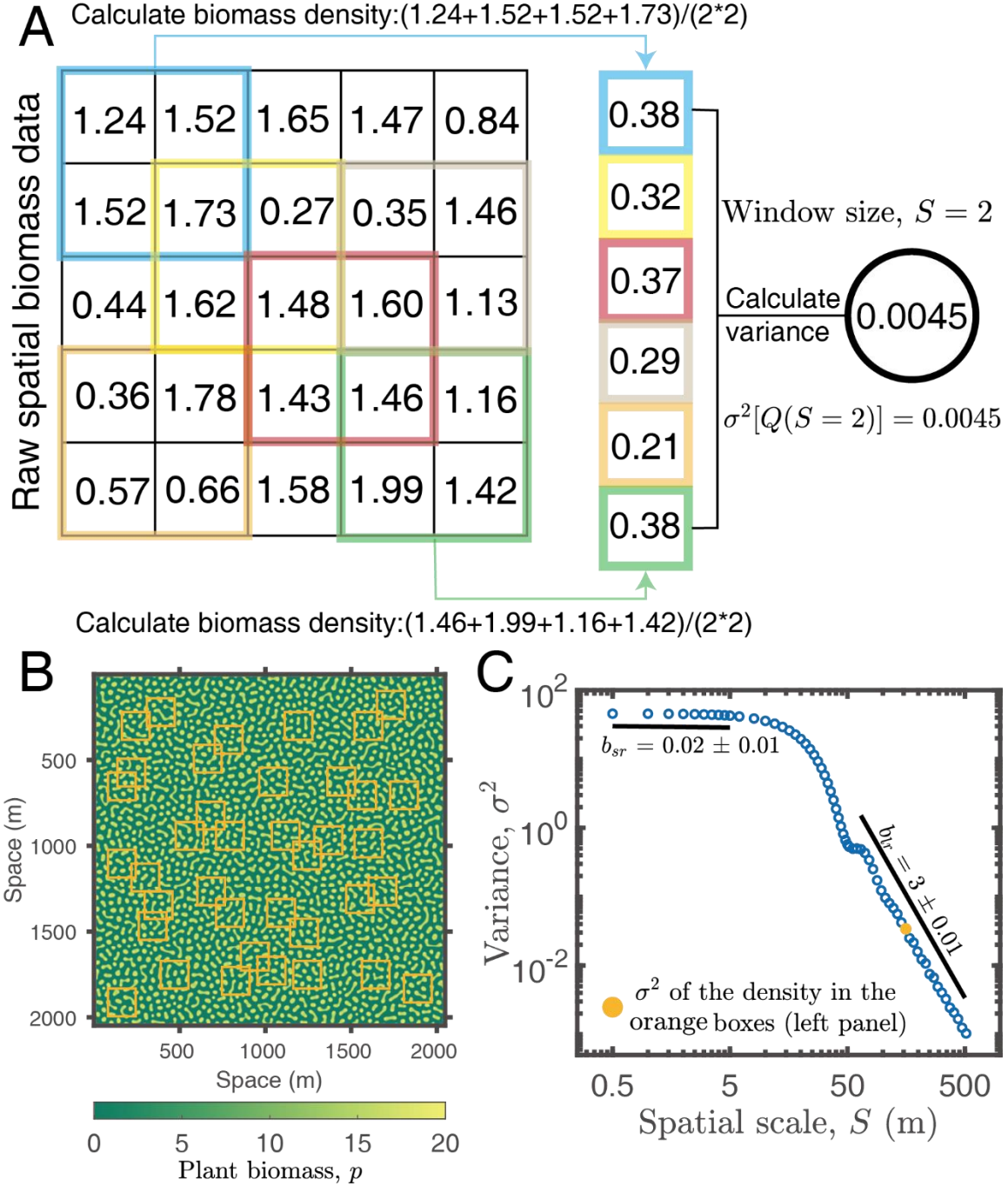


Fig. S9. The schematic diagram of calculating density fluctuation of a spatial pattern with the box throwing method. (A) Calculating the variance of a spatial biomass pattern under a fixed window size of $S = 2$. The 5x5 matrix data is randomly generated. (B) and (C) Calculating the density fluctuation of a spatial pattern. The spatial pattern is from Fig. 1E1 in the main text.

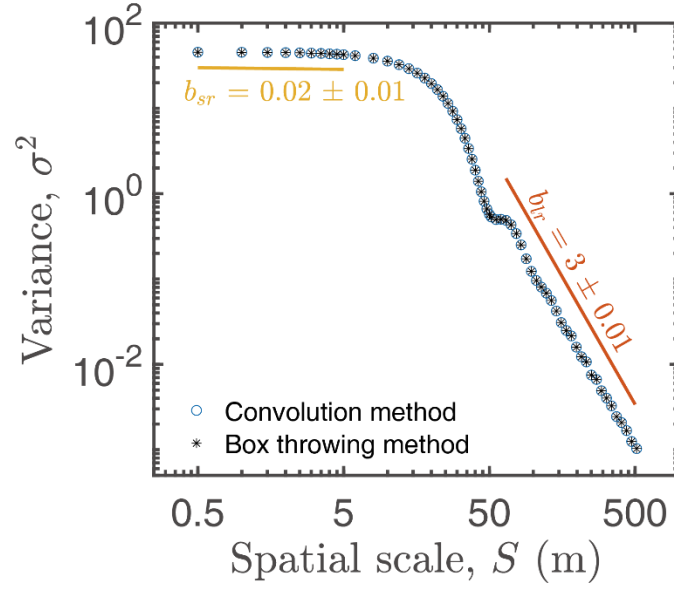


Fig. S10. A comparison between the box-throwing method and the convolution method in calculating the density fluctuation of a spatial pattern. The spatial pattern is from Fig. 1E1 in the main text.

SI References

1. C. A. Klausmeier, Regular and Irregular Patterns in Semi-arid Vegetation. *Science* **284**, 1826–1828 (1999).
2. B. J. Kealy, D. J. Wollkind, A Nonlinear Stability Analysis of Vegetative Turing Pattern Formation for an Interaction–Diffusion Plant–Surface Water Model System in an Arid Flat Environment. *Bull Math Biol* **74**, 803–833 (2012).
3. R. Bastiaansen, *et al.*, Multistability of model and real dryland ecosystems through spatial self-organization. *Proc Natl Acad Sci USA* **115**, 11256–11261 (2018).
4. J. A. Sherratt, Using wavelength and slope to infer the historical origin of semi-arid vegetation bands. *Proc Natl Acad Sci USA* **112**, 4202–4207 (2015).
5. J. von Hardenberg, E. Meron, M. Shachak, Y. Zarmi, Diversity of Vegetation Patterns and Desertification. *Phys. Rev. Lett.* **87**, 198101 (2001).
6. S. Getzin, *et al.*, Discovery of fairy circles in Australia supports self-organization theory. *Proc Natl Acad Sci USA* **113**, 3551–3556 (2016).
7. E. Gilad, J. Von Hardenberg, A. Provenzale, M. Shachak, E. Meron, A mathematical model of plants as ecosystem engineers. *Journal of Theoretical Biology* **244**, 680–691 (2007).
8. V. Dakos, S. Kéfi, M. Rietkerk, E. H. van Nes, M. Scheffer, Slowing Down in Spatially Patterned Ecosystems at the Brink of Collapse. *The American Naturalist* **177**, E153–E166 (2011).
9. G. J. Lord, C. E. Powell, T. Shardlow, *An introduction to computational stochastic PDEs* (Cambridge University Press, 2014).
10. N. Barbier, P. Couteron, J. Lejoly, V. Deblauwe, O. Lejeune, Self-organized vegetation patterning as a fingerprint of climate and human impact on semi-arid ecosystems: Self-organized vegetation patterning. *Journal of Ecology* **94**, 537–547 (2006).